METRIC
DOD-K-24674(SH)
21 August 1986

MILITARY SPECIFICATION

KIT, DISSOLVED OXYGEN (METHOD C), NAVAL SHIPBOARD (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a water testing outfit using an organic dye for the determination of dissolved oxygen in feedwater.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes, Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-P-116 - Preservation, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1193 - Standard Specification for Reagent Water. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Definition. The method C dissolved oxygen test kit measures the amount of dissolved oxygen present in deaerated feedwater at the parts per billion (p/b) level. The kit consists of sampling equipment, clamp, glass sampler, rubber tubing, snapper, test equipment, evacuated ampoules containing test chemicals, and a standard color comparator.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.3).

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3.3 Characteristics.

3.3.1 Performance. The method C dissolved oxygen test kit shall measure the amount of dissolved oxygen in deaerated feedwater having a temperature of 21 to 38 degrees Celsius (°C) (70 to 100 degrees Fahrenheit (°F)) between 0 and 40 p/b. The test results obtained shall be plus or minus 2.5 p/b of the oxygen level when the actual oxygen value is less than 15 p/b and within plus or minus 5 p/b when the actual oxygen value is between 15 and 40 p/b. When the actual oxygen value is greater than 40 p/b the test shall indicate a value of greater than 40 p/b. Feedwater contaminents or treatment chemicals, chlorides and hardness or phosphates, morpoline, EDTA and hydrazine, shall not interfere with the measurement of dissolved oxygen.

3.3.2 Physical characteristics.

3.3.2.1 Component parts. The method C dissolved oxygen test kit shall consist of the following:

(a) One colorimetric comparator with eight liquid color standards representing 0, 5, 10, 15, 20, 25, 30 and 40 p/b dissolved oxygen, respectively
(b) One box of 30 ampoules
(c) One snapper tool
(d) One glass sampler
(e) One finger cot
(f) One utility clamp
(g) One chain type clamp
(h) One piece of 4.8 millimeters (mm) (3/16-inch) inside diameter (id) neoprene tubing, 914.4 mm (3-foot) length
(i) One set of operating instructions
(j) One storage case

3.3.2.1.1 Colorimetric comparator. Details and dimensions of the color comparator used for making the dissolved oxygen determinations shall be as shown on figure 1. The comparator shall consist of a transparent tube in which eight liquid-filled glass vial color standards are mounted and enclosed. There shall be a center space into which the test ampoule can be inserted for color comparison. The comparator shall have the color standards representing 0, 5, 10, 15, 20, 25, 30 and 40 p/b dissolved oxygen. These color standards shall range in color from clear to pink to reddish-pink.

3.3.2.1.2 Test ampoules. Details and dimensions of the test ampoules used in determining the dissolved oxygen content shall be as shown on figure 2. They shall be made of glass, flat at one end and drawn to a tapered hermetically sealed closure at the other end. The tapered end shall be scored so that it breaks easily and evenly when placed in the snapper tool and broken. A color-forming reagent shall be contained in each ampoule. This color-forming reagent shall be clear to pale yellow in color with no sediment present. The ampoules shall contain a vacuum so they will self fill when the tips are broken after they have been placed in the sample to be tested. When filled, the ampoules shall contain the sample and reagent. A small inert gas bubble shall be present in the broken ampoule so that the mixing of the sample and reagent will be facilitated. The ampoules shall be packaged 30 to a box. The details and dimensions

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of the box of ampoules shall be as shown on figures 2 and 3. When tested, they shall meet the requirements specified in 4.5.2.

3.3.2.1.3 Snapper tool. Details and dimensions of the plastic snapper tool used in breaking the tip of the glass measuring ampoules shall be as shown on figure 4. The snapper shall easily and evenly break the scored tip of the measuring ampoule while it is submerged in the sample to be tested. The breaking surface shall not become excessively scored as to prevent the breaking of a box of 30 ampoules.

3.3.2.1.4 Glass sampler. Details and dimensions of the glass sampler shall be as shown on figure 5. The sampler shall easily contain the snapper tool as a flow of water passes upward through it. The tip of the sampler shall accommodate a piece of 4.8 mm (3/16 inch) id tubing.

3.3.2.1.5 Finger cot. The finger cot shall be made from thin rubber, medium size. It shall be used for an entire box of 30 ampoules.

3.3.2.1.6 Utility clamp. The utility clamp shall be as shown on figure 6. It shall be made of corrosion resistant material with adjustable vinyl covered jaws at one end for holding the glass sampler in a vertical position and with an adjustable thumb screw type holder at the other end to fit rods up to 13 mm (1/2 inch) in diameter.

3.3.2.1.7 Chain clamp. The chain clamp shall have a 152.4 mm (6-inch) extension rod that shall attach to a 50.8 to 101.6 mm (2- to 4-inch) diameter pipe. The clamp and rod, as shown on figure 7, shall support the utility clamp and the glass sampler rigidly. The extension rod shall have a 9.5 mm (3/8-inch) diameter.

3.3.2.1.8 Neoprene tubing. The rubber tubing shall be black neoprene rubber tubing with 4.8 mm (3/16 inch) id and at least 1.6 mm (1/16 inch) wall thickness. The length shall be 914.4 mm (3 feet).

3.3.2.1.9 Instructions for use. The following instructions for use shall be included on a piece of 215.9 by 279.4 mm (8-1/2 by 11-inch) paper folded inside the storage case:

"TESTING OUTFIT, FEEDWATER, DISSOLVED OXYGEN CONTENT

1. Apparatus

   (a) Chain clamp
   (b) Utility clamp
   (c) Rubber tubing
   (d) Glass sampler
   (e) Plastic snapper
   (f) Thermometer (0 to 100°C) (32 to 212°F)
   (g) Finger cot

2. Reagents

   (a) Ampoule. An ampoule containing the needed reagents is supplied.

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3. Interferences

(a) Colored water or suspended matter in the sample may interfere in detecting comparator color differences.

4. Procedure

(a) Attach one end of a short length of black neoprene rubber tubing to the outlet nipple of the sample cooler.
(b) Attach the glass sampling tube to the other end of the rubber tubing and secure the glass tube upright with the utility clamp. The utility clamp and the chain clamp shall be permanently attached near the sample cooler.
(c) With the cooling water flow secured (see notes), start a flow of sample water at a maximum rate to flush the sample system.

NOTES:
(1) When the sample cooler uses sea water or drinking water as coolant, close the cooling water inlet valve to the cooler in preparation for hot flushing. To preclude cooler rupture during hot flushing, a cooling water outlet valve has not been installed in these systems.
(2) When the sample cooler uses chilled water from the air conditioning main as coolant, close only the cooling water outlet valve for the sample cooler in preparation for hot flushing. Cooling water flow should be controlled by the cooling water outlet valve only. All other cooling water valves remain open.

(d) Initiate coolant flow to flush the line with sample flow at a temperature of 60 to 93°C (140 to 200°F) for at least 5 minutes. Knead the rubber tubing to dislodge air bubbles.
(e) Adjust cooling water and sample flow rate to obtain a temperature of between 21 and 38°C (70 and 100°F) as measured by a thermometer.
(f) Place the snapper in the sampling tube.
(g) Place an ampoule in the snapper. Sample flow must be sufficient to float the ampoule without forcing it out of the snapper. Hold the ampoule down for 1 minute.
(h) Press down on the ampoule to break the tip. A sample will be drawn in and a small bubble of inert gas will remain in the ampoule.

NOTE: The test must be run immediately after collection since the sample will absorb oxygen from the atmosphere.

(i) Remove the ampoule from the snapper, immediately cover the open end with a finger (the finger should be protected with a finger cot), and invert the ampoule 10 times allowing the gas bubble to transverse the tube to mix reagent and sample.
(j) Wipe the ampoule dry and place it in the center of the comparator, with the open end up.
(k) View the comparator from the bottom with the top pointing at a daylight fluorescent lamp.

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(l) Rotate the comparator until a color match is obtained or the sample color lies between two of the standard colors or the sample color is darker than the 40 micrograms per liter (µg/L) (40 p/b) standard.

(1) If a color match is obtained, record that value in the log.
(2) If the color lies between the two standard colors, add the two values, divide by two, and record that value in the log.
(3) If the color of the sample is darker than the 40 µg/L (40 p/b) standard, record greater than 40 µg/L (40 p/b) in the log.

(m) Dispose of the used ampoule in such a manner that the broken glass does not present a safety hazard to personnel.
(n) After finishing the test, secure the feedwater flow, secure the coolant and disconnect the rubber tubing and sampling tube.

5. Range of error

(a) The range of error of the test is plus or minus 2.5 µg/L (2.5 p/b) dissolved oxygen."

3.3.2.1.10 Storage case. Details and dimensions of the storage case for the dissolved oxygen test kit shall be as shown on figure 8. The case shall be molded plastic, one piece, with the upper and lower portion joined by a thin plastic hinge. The upper half shall contain a rectangular cavity into which a piece of foam or foam-like packing material shall be inserted. This packing material shall have cutouts for storage of the chain-type clamp and utility clamp. The lower half of the case shall also have a rectangular cavity into which a piece of foam or foam-like material shall be inserted. This material shall have cutouts for storing a box of 30 ampoules, one snapper tool, one glass sampler, one finger cot, one comparator and one piece of tubing. A snap-type locking device shall be on the upper and lower halves, with the hinged snap on the upper half and the catch on the bottom half.

3.3.3 Color standards. The color standards shall be fabricated to the dimensions shown on figure 1. The comparator shall have the date of manufacture stamped on it. The pink and reddish pink colors shall yield the results as specified in table I when tested in accordance with 4.5.1.

TABLE I. Dissolved oxygen solution values for colorimetric comparator.

Standard (p/b dissolved oxygen)	Accepted values (p/b)
0	0
5	5 ± 2.5
10	10 ± 2.5
15	15 ± 5.0
20	20 ± 5.0
25	25 ± 5.0
30	30 ± 5.0
40	35, 40, greater than 40

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3.3.4 Color standard stability. The colorant used in the manufacture of the color standards for this equipment shall exhibit in the completed standards a stability that shall give the specified values of table I as determined at the end of typical shipboard use. The effective shelf life of the colorimetric comparator shall be 2 years from the date of manufacture. The procedures for determining the accuracy of the comparator shall be as specified in 4.5.1.

3.3.5 Ampoule stability. Each box of ampoules shall have the month and year of manufacture stamped on it. The effective shelf life of a box of ampoules shall be 24 months from the date of manufacture. The contractor shall show that the dissolved oxygen values obtained are as specified by using the procedures of 4.5.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall consist of the examination and tests shown in table II. Unless otherwise specified in the contract or purchase order (see 6.2), the Naval Ship Systems Engineering Station (NAVSSES) Code 053D, Philadelphia, PA 19112-5083 is responsible for the performance of all inspection requirements.

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TABLE II. First article inspection.

Inspection	Requirement	Examination and test
Visual and dimensional examination	3.3.2.1.1 to 3.3.2.1.10	4.4.1
Comparator	3.3.2.1.1	4.5.1
Ampoule	3.3.2.1.2	4.5.2

4.2.2 Quality conformance inspection. Quality conformance inspection shall consist of the visual and dimensional examination of table II, the ampoule test of 4.5.2, and the comparator test of 4.5.1 selected in accordance with table III from each inspection lot.

4.3 Sampling.

4.3.1 Inspection lot. For purposes of sampling, a lot shall consist of all complete testing outfits or component parts presented at one time.

4.3.2 Sampling for examination. A random sample of complete testing outfits or component parts shall be selected in accordance with table III from each inspection lot offered for Government inspection of visual and dimensional characteristics. Sampling shall be in accordance with MIL-STD-105 at an acceptable quality level (AQL) of 2.5 percent defective.

TABLE III. Sampling for examination.

Number of units or component parts in inspection lot	Number of outfits or component parts in sample	Acceptance number (defectives)	Rejection number (defectives)
25 or under	5	0	1
26 - 50	8	0	1
51 - 90	13	1	2
91 - 150	20	1	2
151 - 280	32	2	3
281 - 500	50	3	4
501 - 1200	80	5	6

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4.4 Examination.

4.4.1 Visual and dimensional examination. Each of the complete testing outfits or component parts selected in accordance with table III shall be visually and dimensionally examined to verify conformance with this specification. If any complete testing outfit or component part in the sample contains one or more visual or dimensional defects, this shall be cause for rejection of the defective part or outfit. If the number of defective testing outfits or component parts in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.5 Test procedures.

4.5.1 Comparator test. The following test procedure shall be used to determine the accuracy of the comparator.

4.5.1.1 Reagent. Reagent water shall conform to type II of ASTM D 1193.

4.5.1.2 Apparatus. The apparatus shall include the following:

(a) Hellige 0 to 40 µg/L (0 to 40 p/b) dissolved oxygen color wheel or equal.
(b) Hellige standard comparator or equal.
(c) Two 40 mm (1-9/16 inch) drawn viewing tubes or equal.
(d) Daylite comparator illuminator or equal.
(e) Comparator test solutions representing the table I dissolved oxygen values available from the contractor.

4.5.1.3 Procedure.

(a) Place standard comparator on comparator illuminator.
(b) Open front cover of standard comparator and insert color wheel with the number plates facing upward.
(c) Fill one 40 mm (1-9/16 inch) viewing tube with distilled water.
(d) Place this tube in the left side opening after turning spring clip aside.
(e) Fill one 40 mm (1-9/16 inch) viewing tube with the first dissolved oxygen test comparator solution.
(f) Place this tube in the right side opening after turning spring clip aside.
(g) Turn on comparator illuminator.
(h) View both solutions through the lens of the magnifying prism attachment.
(i) Rotate the color wheel until a color match is obtained with one of the standards. Read the result directly from the figure seen in the upper right hand corner. If the color of the test solution is intermediate between two glass standards, add the value of the two standards and divide by 2.
(j) Record the result.
(k) Repeat steps (e) through (j) for the seven remaining dissolved oxygen test comparator solutions.
(l) Values for the test solutions shall correspond to those in table I.

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4.5.2 Ampoule test. The following test procedures shall be used to determine the accuracy of the ampoules.

4.5.2.1 Apparatus. The apparatus shall include the following:

(a) Deaerating feed tank (DFT) capable of lowering the dissolved oxygen content of water to 5 µg/L (5 p/b).
(b) Dissolved oxygen analyzer (manufactured by Rexnord, Inc., Malvern, PA., or equal).
(c) Method C dissolved oxygen test kit.

4.5.2.2 Procedure.

(a) Light off DFT.
(b) When DFT comes on line, start up dissolved oxygen analyzer.
(c) Set up method C dissolved oxygen test kit.
(d) When the dissolved oxygen level of the DFT, as measured by the dissolved oxygen meter, measures 15 µg/L (15 p/b), snap one test ampoule.
(e) After testing is completed at the 15 µg/L (15 p/b) level, lower the dissolved oxygen output level of the DFT to 10 µg/L (10 p/b).
(f) When the dissolved oxygen level of the DFT, as measured by the dissolved oxygen meter, measures 10 µg/L (10 p/b), snap another test ampoule.
(g) Compare ampoules to dissolved oxygen test kit comparator. Values obtained shall be within the limits specified in table I for the two values 10 and 15 µg/L (10 and 15 p/b) dissolved oxygen.

4.6 Packaging inspection. Sample packages and packs and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation. Preservation shall be level A, C, or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Test kit. Each test kit with carrying case (see 3.3.2.1.10) shall be individually preserved in accordance with method III of MIL-P-116 and unit packed in paperboard set up or metal stayed boxes or fiberboard boxes conforming to PPP-B-676, PPP-B-665, or PPP-B-636, respectively. Boxes shall conform to the weather resistant class or grade as applicable with type, variety and style selection at the option of the contractor. Box closure shall be in accordance with the applicable box specification or appendix thereto, with method V closure applicable to PPP-B-636 boxes.

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5.1.1.2 Operating space items. Comparator and dissolved oxygen ampoules to be furnished as replacement items shall be preserved in accordance with method III of MIL-P-116 in individual paperboard boxes provided with partitions or separators, or individually wrapped to prevent contact or damage to the comparator or ampoules. Paperboard boxes shall conform to any style, class, or grade of PPP-B-566 or PPP-B-676. Separator-divider material shall be of material equal to the box material.

5.1.2 Level C. Test kits and operating space items shall be preserved as specified in 5.1.1.1 and 5.1.1.2 except that non-weather resistant domestic class or grade containers (boxes) may be used.

5.1.3 Commercial. Commercial preservation shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be levels B, C, or commercial, as specified (see 6.2).

5.2.1 Level B. Test kits and operating space items, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, class weather-resistant, or class 2 of PPP-B-640. Boxes shall be closed and waterproofed in accordance with the applicable box specification or appendix thereto with method V closure applicable to PPP-B-636 boxes. Boxes shall be reinforced with pressure sensitive reinforced tape or non-metallic banding in lieu of steel strapping.

5.2.2 Level C. Test kits and operating space items preserved as specified in 5.1 shall be packed in boxes as specified in 5.2.1 except that the domestic class or grade shall apply, and waterproofing is not required.

5.2.3 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Levels A, B, and C. In addition to any special marking required herein or by the contract or order (see 6.2) each test kit container (see 3.3.2.1), operating space item pack (see 5.1.1.2), interior packs, and exterior shipping containers shall be marked in accordance with MIL-STD-129 and shall include the following:

(a) National stock number.
(b) Name and class of item.
(c) Manufacturer's or contractor's name or both.
(d) Contract number.
(e) Quantity or net content.
(f) Date of manufacture: month and year.
(g) Expiration date of 12 months from date of manufacture for the colorimetric comparator. Expiration date of 24 months from date of manufacture for the ampoules.
(h) Note: "Store ampoules in the dark. Ampoules deteriorate upon exposure to light. Mandatory shelf life material. Discard material after expiration date."
(i) Lot or batch number.

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5.3.2 Commercial. In addition to any special marking required herein or in the contract or order (see 6.2), commercial marking shall be as specified in ASTM D 3951.

5.3.3 Special marking. Shelf life, use, and warning labels as applicable for the comparator ampoules and dissolved oxygen ampoules shall be in accordance with MIL-STD-129. Bar code markings of MIL-STD-129 shall apply for interior and exterior containers.

6. NOTES

6.1 Intended use. This test kit is to be used for the determination of dissolved oxygen in feedwater.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) When first article tests are required for inspection and approval (see 3.2, 4.2.1 and 6.3).
(c) If quality conformance tests are required (see 4.2.2).
(d) Levels of preservation and packing (see 5.1 and 5.2).
(e) Commercial preservation and packing required (see 5.1.3 and 5.2.3).
(f) Special marking required (see 5.3.1, 5.3.2, and 5.3.3).
(g) Commercial marking required (see 5.3.2).
(h) Patent indemnity clause 52.227-3, alternate II of the FAR, Federal Acquisition Regulation (see 6.4).

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1 Bid data. Bidders offering individual components rather than complete kits will be subject to the following special requirements:

(a) Bidders offering items of apparatus only will not be required to perform the tests of 4.5.1 and 4.5.2, but will be required to fulfill all other requirements of this specification. These bidders may request that the Government waive the requirements for first article samples, provided the bidder is offering a product which has previously been acquired or tested by the Government, and that bidders offering such products, who wish to rely on such productions or tests must furnish evidence with the bid that prior Government approval is presently appropriate for the pending acquisition.

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(b) Bidders offering colorimetric comparator or ampoule items only will be required to fulfill all the requirements and tests of this specification. These bidders may request that the Government waive the requirements for first article samples, provided the bidder is offering a product which has previously been acquired or tested by the Government, and that bidders offering such products, who wish to rely on such productions or tests must furnish evidence with the bid that prior Government approval is presently appropriate for the pending acquisition.

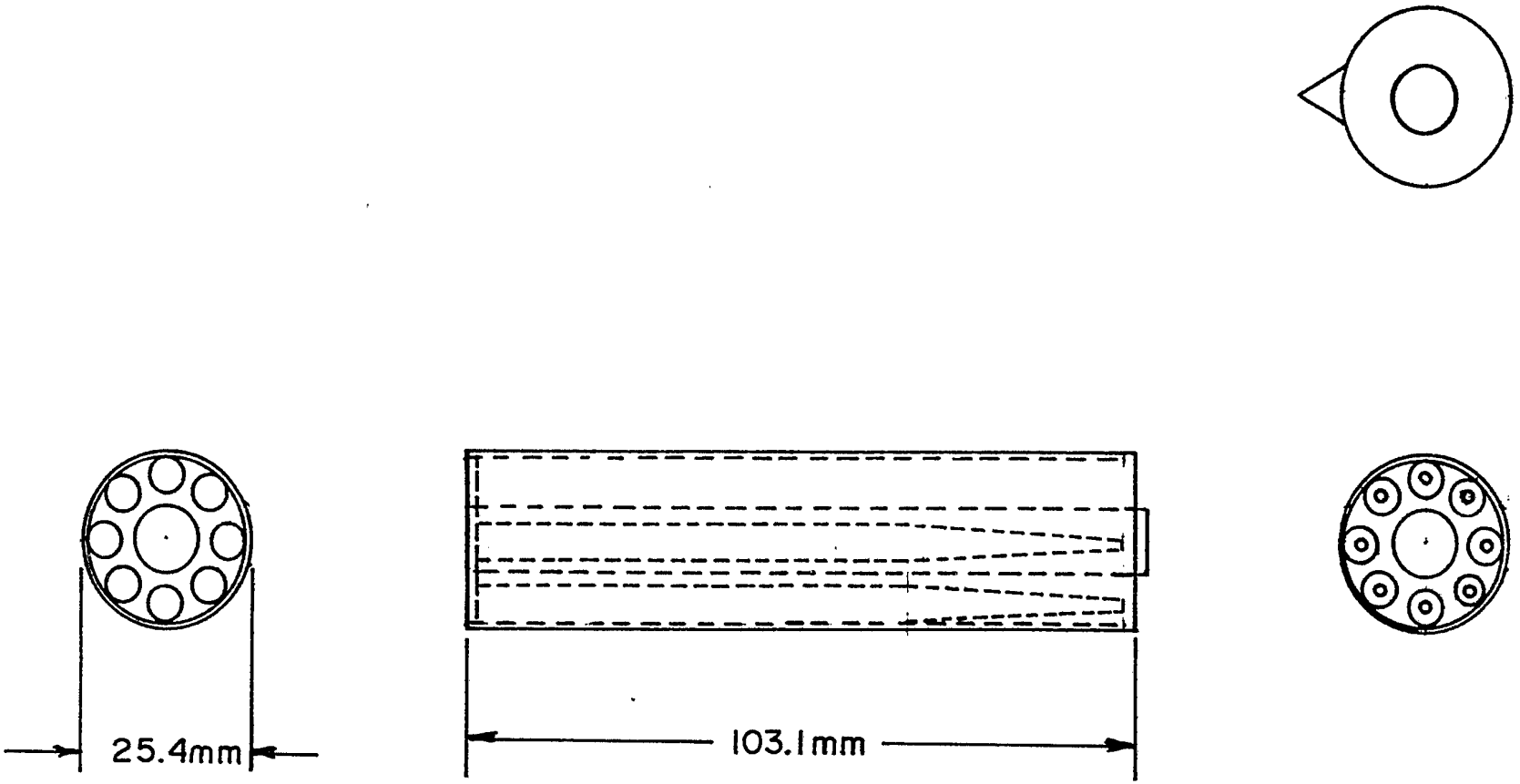
6.4 Patent notice. The Government does not have a royalty-free license for U.S. Patent 3,634,038, Device for the Quantitative Colorimetric Analysis of Fluids, patented January 11, 1972.

6.5 Subject term (key word) listing.

Ampoule
Color comparator
Color standards
Deaerated feedwater
Dissolved oxygen
Organic dye
Oxygen test kit
Reagent water

Preparing activity:
Navy - SH
(Project 6630-N420)

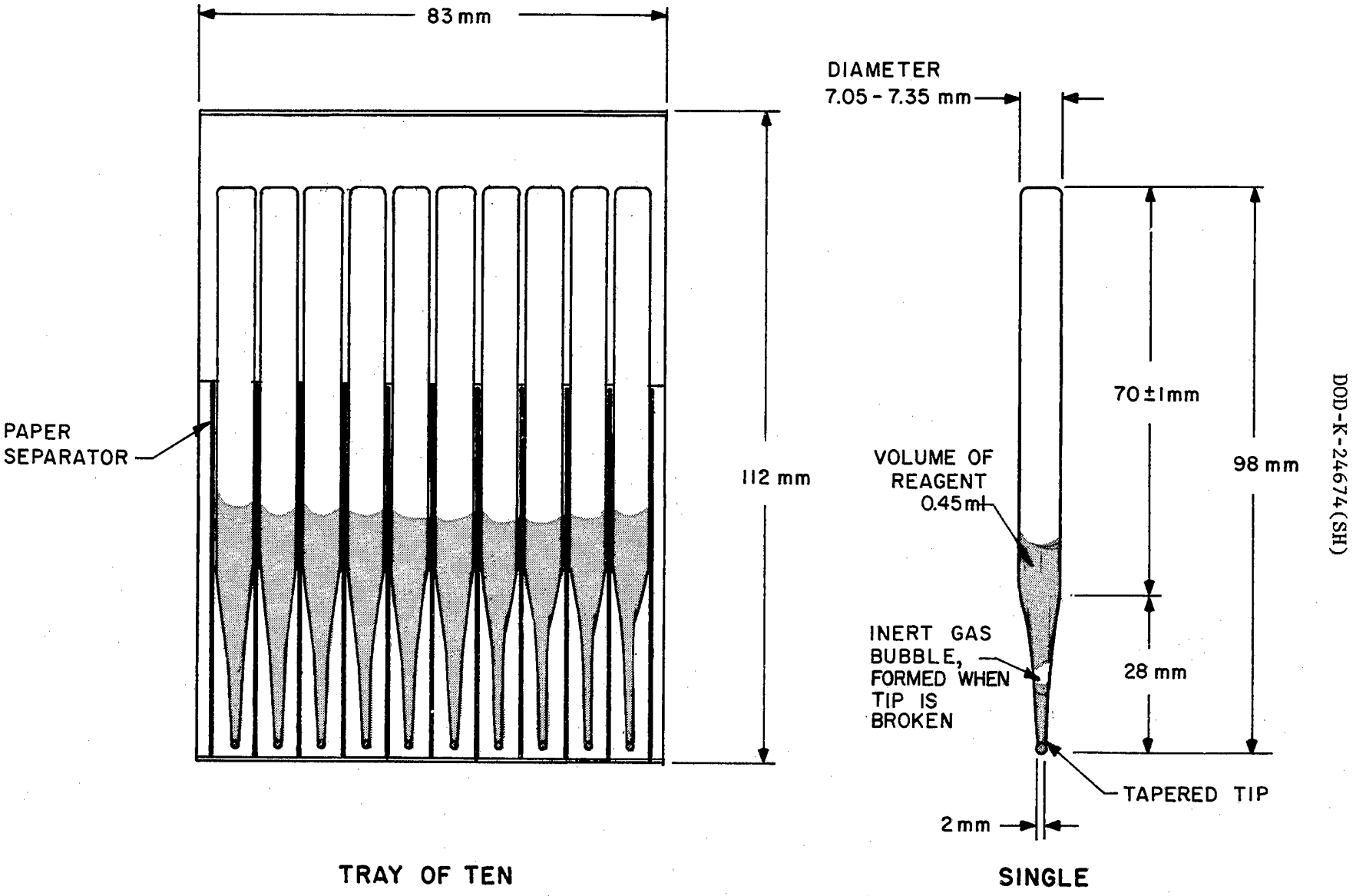
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FIGURE 1. Comparator.

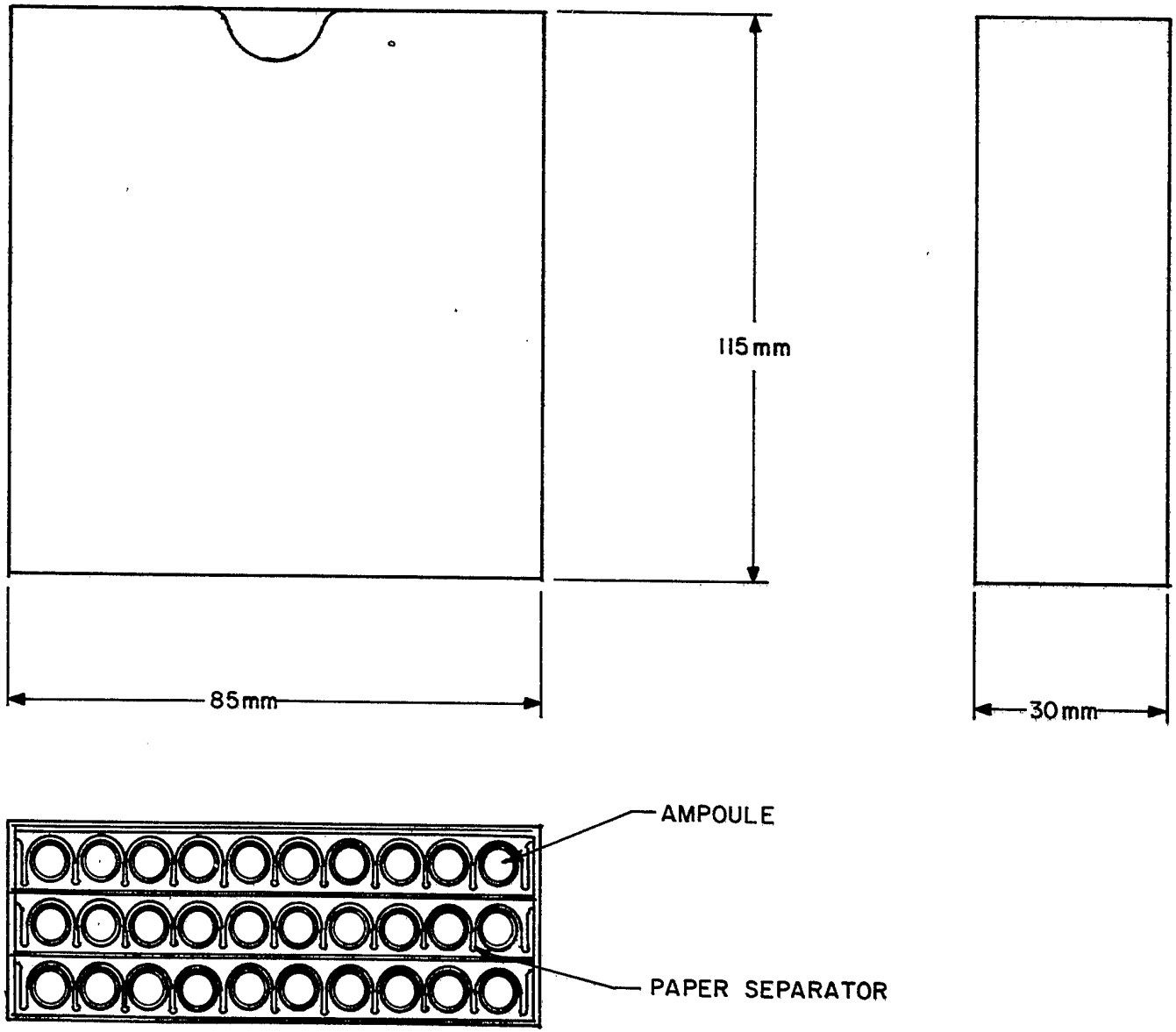
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SH 131838

FIGURE 2. Glass ampoules.

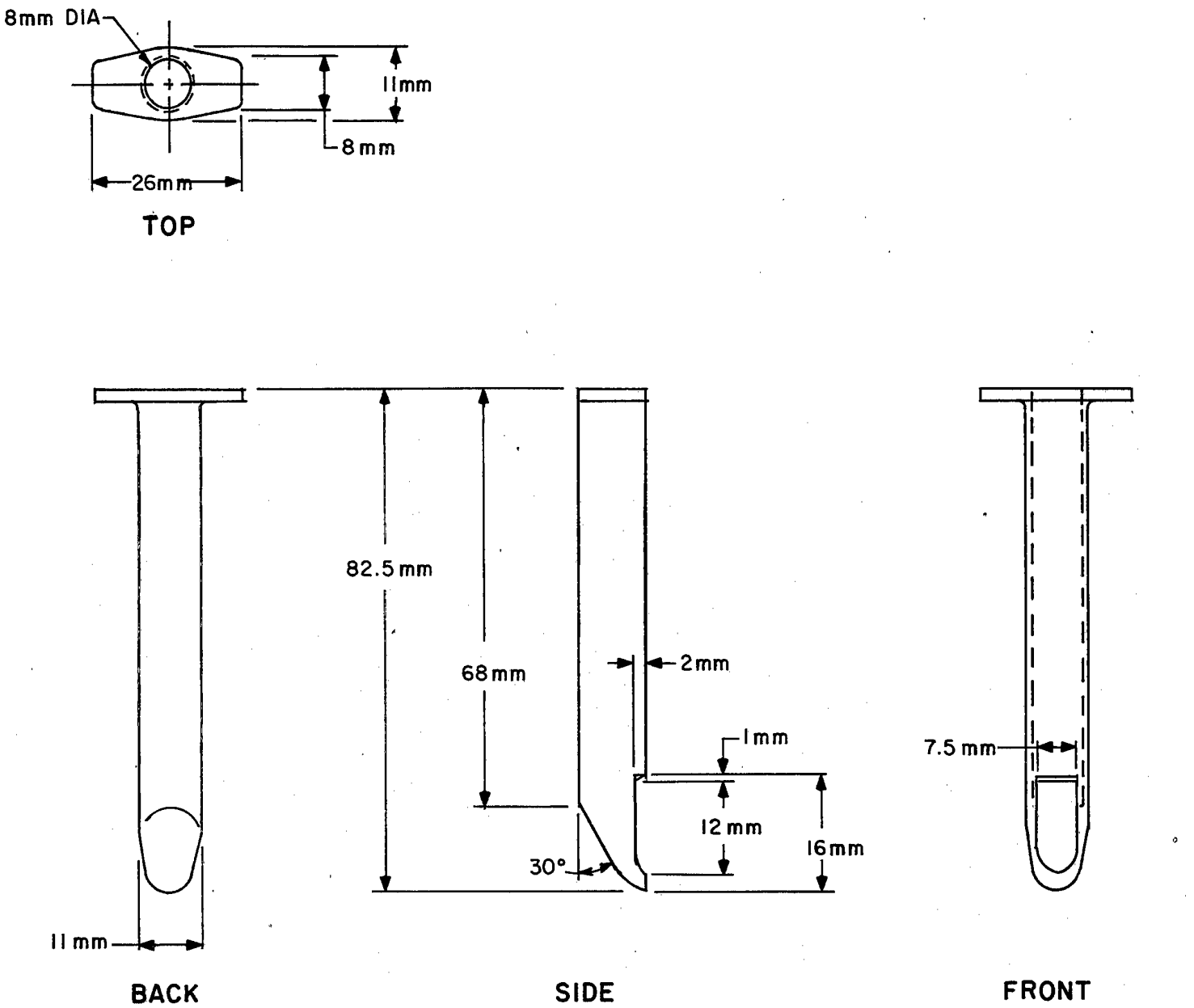
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SH 131839

FIGURE 3. Box of ampoules.

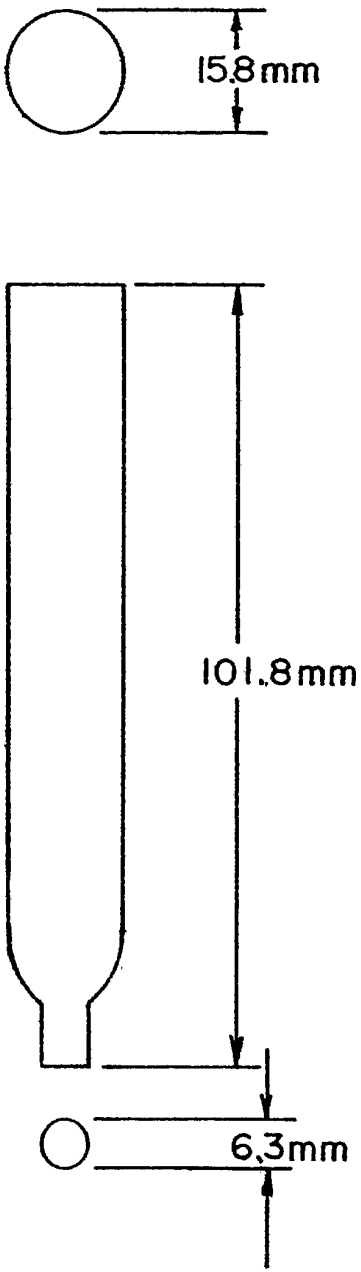
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SH 131840

FIGURE 4. Snapper.

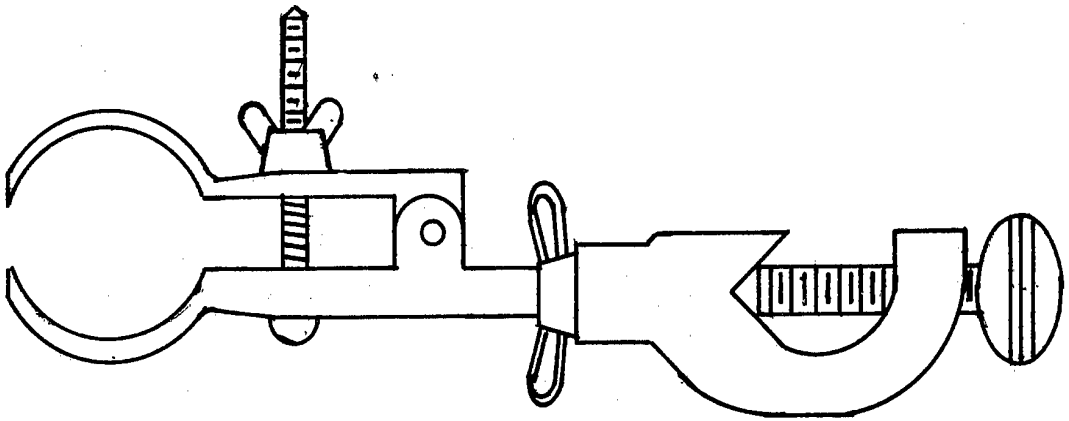
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SH 131841

FIGURE 5. Glass sampler.

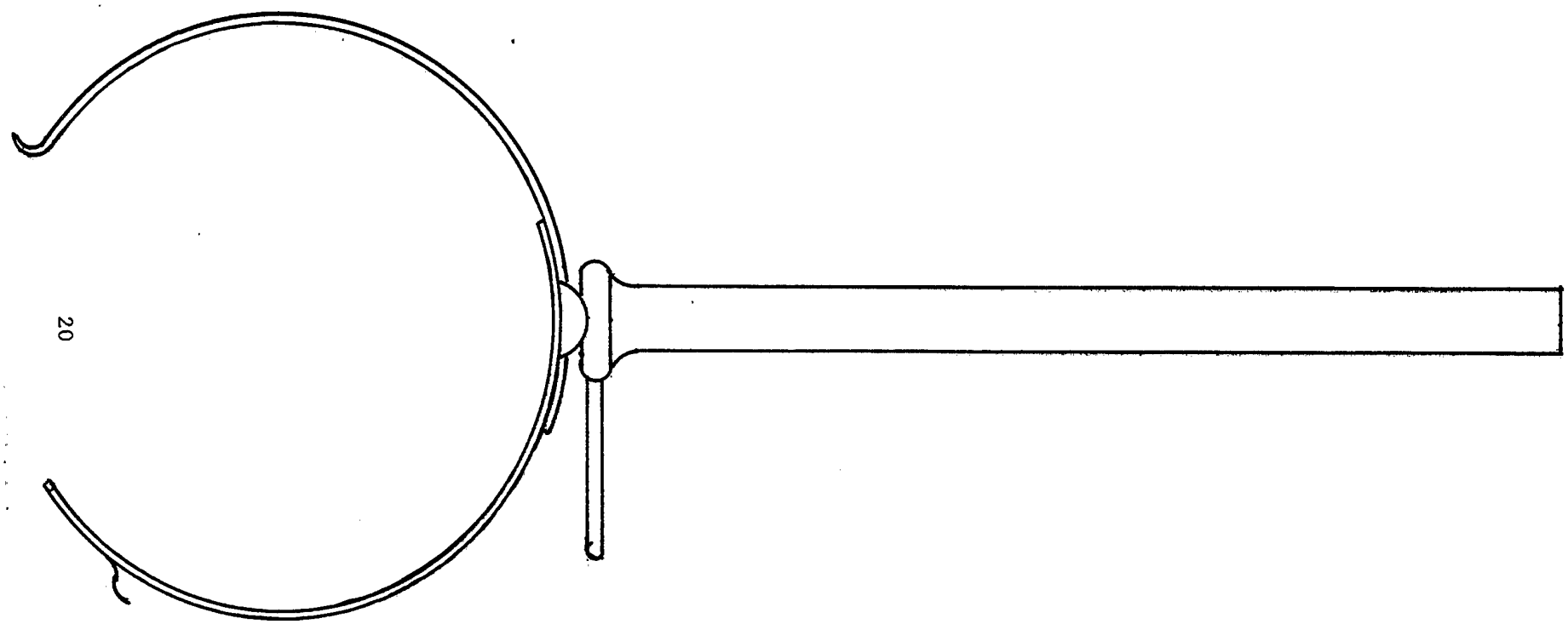
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SH 131842

FIGURE 6. Utility clamp.

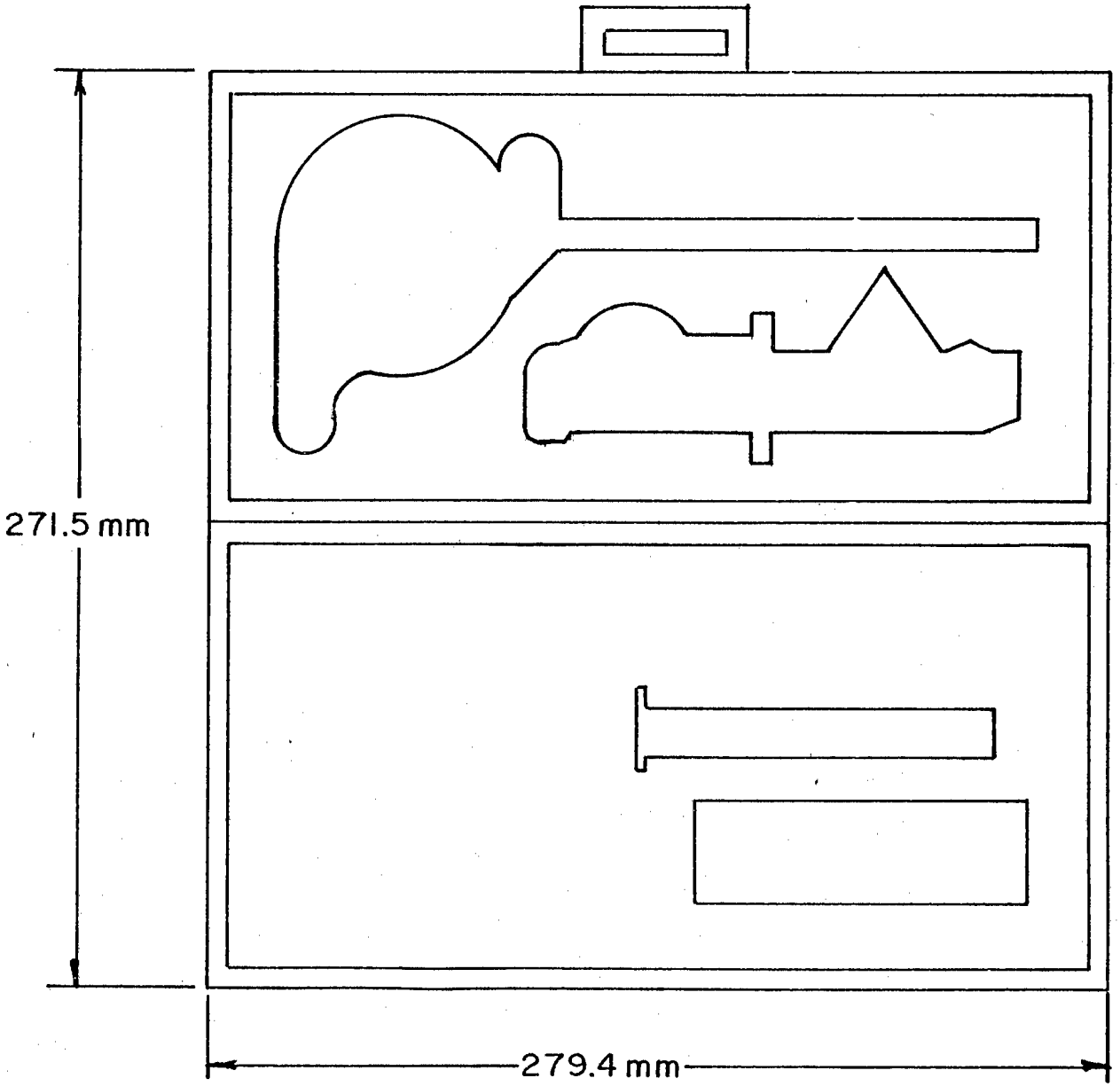
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SH 131843

FIGURE 7. Chain clamp.

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SH 131844

1/2 Scale

FIGURE 8. Case.

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